Research.gov - Proposal Review Page

research.gov/gapps-web/gapps/viewApplicationReview

Agency Name:

National Science Foundation

Agency Tracking Number:

2533187

Organization:

Translational Impacts

NSF Program:

SafeOSE

PI/PD:

Millman, Kenneth

Application Title:

NSF Safe-OSE: Securing the Scientific Python Ecosystem

Rating:

Good

Review

Summary

Proposal Synopsis

The goal of this proposal is to address the vulnerabilities in safety, security, and privacy in the Scientific Python Open Source Ecosystem (SP-OSE). The team proposes a community-backed coordination process, (Scientific Python Ecosystem Coordination (SPEC), to improve design of an interconnected de-centralized open-source ecosystem and disseminate ecosystem-wide security standards and best practices and enhance the integrity and reliability of scientific research across numerous domains.

Intellectual Merit

Strengths

+The proposal addresses a widely used open-source tool and programming language; improvements to the system would benefit large communities ranging from physics, biosciences, education, etc. Potential for impact and intellectual contributions and streamlining scientific research in these fields is very strong.

+The team presents strong preliminary data and evidence of community-driven approach through

+The PI funded the Scientific Python Project (SPP) which brings together communities and provides a strong foundation for the proposed coordination network, with over 80 team members, 29 affiliated Python libraries, and many subscribers involved. This forum has a well-defined management structure and hosts regular meetings. Several project are already ongoing.

+ The broad range of scientific domain with increase accessibility.

+ The model addresses several classes of vulnerabilities.

+The team is qualified and has extensive experience in open source programming.

+ There is adequate access to resources.

Weaknesses

-The project activities focus primarily on the design of an existing system - SP-OSE, and thus improvements are incremental rather than transformational.

- Technical aspects of the proposed model are very general. It is unclear what data will be used to develop community standards and ecosystem-wide protocol to guide the proposed improvements and their implementation in existing workflows. Details are missing on how the model will be developed, tested and implemented.

- It is also unclear how many users will provide feedback and how this feedback will be implemented.

- How the accuracy of data will be ensured and validated is unclear.

- Success criteria and milestones are missing.

Broader Impacts

Strengths

+The project has the potential to deliver a vastly improved SP-OSE systems that would streamline security features and protect infrastructure and sensitive data that rely on Python.

+The new set of standards for open source security can be widely implemented across many open source communities.

Weaknesses
-It is unclear how the proposed improvements will be disseminated the broader community and how other users and institutions gain access to the new model.

Additional Solicitation-specific Review Criteria
Strengths & Weaknesses
+The proposed model to enhanced security features scientific and technological innovations across different field is essential to enhance security of scientific data

+The proposal clearly describes the vulnerabilities that will be addressed.

+ There is an external advisory board in place.

+A security Working Group Oversight will be established with regular meetings to evaluate progress

+The project builds upon an existing successful network, institutional partnerships and efforts that already engage a broad community to provide feedback and support the development efforts

+There are multiple letters from partner organization attesting the importance of the vulnerabilities addressed, and providing support and collaboration.

-The is little discussion of the origin and dimensionality of data.

Summary Statement
A proposal to enhance security features and address a series of vulnerabilities of Python. The project is well designed and builds upon a strong existing infrastructure and community engagement. The plan will generate novel security features of broad applicability in the scientific Python ecosystem and high potential for adoption across multiple scientific fields. The main weaknesses of this proposal is the lack of details regarding the data source and accuracy, and the insufficient information on how the model will be tested and validated for broader use.